



Deutsche
Akkreditierungsstelle
D-ML-20488-01-00

ArminLabs GmbH · Zirbelstr. 58 2nd floor · 86154 Augsburg · Germany

ArminLabs GmbH
Zirbelstr. 58

D 86154 Augsburg

Patient : **Messineo, Maria**

Date of Birth: 05/13/1970 female

Final report

Order-ID : **839248860**

Page **1/ 2**

Date of Reception/Report : 02/23/2018

02/27/2018

Analysis	Result	Units	Reference Range	Chart
Borrelia EliSpot *				
1 Borrelia b. Full Antigen	!	2 SI		
0-1 = negative				
2-3 = weak positive				
> 3 = positive				
1 Borrelia b. OSP-Mix		1 SI		
0-1 = negative				
2-3 = weak positive				
> 3 = positive				
1 Borrelia burgdorferi LFA-1		1 SI		
0-1 = negative				
2-3 = weak positive				
> 3 = positive				

The results of the Elispot tests indicate weak current cellular activities against *Borrelia burgdorferi*.

Explanation of antigens:

Borrelia-burgdorferi Full Antigen: *Borrelia burgdorferi* B31 reference strain (*Borrelia b sensu stricto*)

Borrelia-burgdorferi Peptide Mix: OspA from *Borrelia b. sensu stricto*, *Borrelia afzelii*, *Borrelia garinii* + OspC native + DbpA recombinant.

Borrelia-burgdorferi LFA-1 (Lymphocyte Function Antigen 1) Own body protein + *Borrelia burgdorferi sensu stricto* (shared epitope). LFA1 can be associated with autoimmune diseases: collagenosis, Rheumatoid Arthritis, vasculitis. If positive or borderline positive look at: ANA, CCP-antibodies, ANCA.

Borrelia IgG-/IgM-SeraSpot®

4 Borr. burgdorferi SeraSpot IgG	positive	negative
4 Borr. SeraSpot VlsE (afzelii)	positive	negative



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Analysis	Result	Units	Reference	Range	Chart
4 Borr. SeraSpot p39 (afzelii)	negative		negative		
4 Borr. SeraSpot p58 (garinii)	negative		negative		
4 Borr. SeraSpot p100 (garinii)	negative		negative		
4 Borr. SeraSpot OspC (afzelii)	positive		negative		
4 Borr. SeraSpot OspC (garinii)	positive		negative		
4 Borr. SeraSpot OspC (s.s.)	positive		negative		
4 Borr. SeraSpot dbpA (afzelii)	negative		negative		
4 Borr. SeraSpot dbpA (garinii)	negative		negative		
4 Borr. SeraSpot dbpA (s.s.)	negative		negative		
4 Borr. burgdorferi SeraSpot IgM	positive		negative		
4 Borr. SeraSpot VlsE (afzelii)	negative		negative		
4 Borr. SeraSpot p39 (afzelii)	negative		negative		
4 Borr. SeraSpot p58 (garinii)	negative		negative		
4 Borr. SeraSpot p100 (garinii)	negative		negative		
4 Borr. SeraSpot OspC (afzelii)	positive		negative		
4 Borr. SeraSpot OspC (garinii)	negative		negative		
4 Borr. SeraSpot OspC (s.s.)	positive		negative		
4 Borr. SeraSpot dbpA (afzelii)	negative		negative		
4 Borr. SeraSpot dbpA (garinii)	negative		negative		
4 Borr. SeraSpot dbpA (s.s.)	negative		negative		

The positive Borrelia burgdorferi-IgG/IgM-antibodies by SeraSpot (modern Borrelia Westernblot) indicate humoral immune responses against Borrelia burgdorferi.

Please look at the Borrelia-EliSpot and the CD57-positive NK-cells. Take into consideration the clinical symptoms and the differential diagnosis (co-infections).

Babesia IgG-/IgM-antibodies

4 Babesia microti IgG-AB (IFT)	<1:16	< 1:16	[....*...]
4 Babesia microti IgM-AB (IFT)	<1:20	< 1:20	[....*...]

Serological no evidence for an infection with Babesia microti.

We recommend to look at the current cellular activity by the Babesia-EliSpot.

validated by

Dr.Armin Schwarzbach